

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 20/05/2019

Day: Monday

Time: 10:00 am to 10:30 am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : B.Optomtry – May 2019
Semester: 3 rd
Section : I
Course Code : BOP303
Course Name: OPTOMETRIC OPTICS-I

Important Instructions:

1. There are two sections in the paper.
2. **SECTION –I** contains 2 Questions:
 - a. MCQs for 20 marks answered in the question paper itself, should be submitted within 30 minutes.
 - b. Short answers for 20 marks (to be answered in separate answer book)
3. **SECTION – II** contains only 2 Question of Short notes and Essays (to be answered in answer book)
4. Marks on the right side suggest the total marks of that question.
5. Maximum time allotted for both section is 3 hours.
6. Candidates who complete **SECTION – I** (MCQs) earlier than 30 minutes can collect main question paper immediately after handing over the **SECTION-I (MCQs)** question paper with answers.
7. Non-programmable calculators are only allowed.
8. Draw the figures where necessary.

SECTION-I

Q.1.	Multiple Choice Questions: (MCQs) Encircle the correct answer	(1 X 10= 10)
1.	What does the root word phot means?	
	a. Colour	
	b. Light	
	c. Disease	
	d. Image	
2.	Pin-cushion distortion is caused by_____	
	a. Minus lens	
	b. Plus lens	
	c. Prism	
	d. Glass slab	

3.	1 degree is equal to ___ prism diopter.	
	a. 1PD	
	b. 2PD	
	c. 1.75PD	
	d. 3PD	
4.	Catoptric power means___	
	a. Image formation with the help of reflection	
	b. Image formation with the help of refraction	
	c. Image formation with the help of reflection and refraction	
	d. None of the above	
5.	Following are the properties of prism, except	
	a. Refraction	
	b. Deviation	
	c. Dispersion	
	d. Distortion	
6.	Given a lens for which $F1=+7.00D$, $F2= -7.00D$, $t=2.0mm$, and $n=1.523$. What is the Approximate power?	
	a. $+14.00D$	
	b. $-14.00D$	
	c. $+7.00D$	
	d. None of the above	
7.	A lens of $+5.00D$ having focal length of___	
	a. 2m	
	b. 2cm	
	c. 20m	
	d. 20cm	
8.	In ophthalmic practice prism is used for_____	
	a. Correction of refractive error	
	b. For treatment of squint	
	c. For vision therapy	
	d. All of the above	
9.	Fresnel prism is manufactured by a process of_____	
	a. Injection molding	
	b. Compression molding	
	c. a&b	
	d. None	
10.	Front vertex power is also known as__	
	a. Neutralizing power	
	b. Equivalent power	
	c. Approximate power	
	d. None of the above	

Identify the sentence as True or False:		(5 × 1 = 5)
11.	A spherical aberration is an off-axis aberration.	
12.	Spherometer is used to check equivalent power of spectacle lens.	
13.	Distance from spectacle plane to back surface of cornea is called vertex distance.	
14.	Spherical equivalent for the prescription of +1.00/+2.50×10 is +1.25Ds	
15.	Thickness of Fresnel prism is about 7mm.	

Match the following:		(5 × 1 = 5)	
16.	1.525	A.	Lens dial
17.	Measuring phoria & fusional vergence	B.	Spherical aberration
18.	VIBGYOR	C.	Rotary prism
19.	Lens Stop	D.	Dispersion
20.	1.530	E.	Fresnel prism

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
3rd Semester of B.Optomety Examination
May 2019

BOP303: Optometric Optics-I

Date: 20/05/2019 Day: Monday Time: 10:30am to 1:00pm Maximum Marks: 50

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SECTION-I

Q.2.	Short answers (10 out of 11)	(10 × 2 = 20)
a.	Define Base curve and Cross curve	
b.	A +5.00D lens before the right eye is decentered 2mm nasal ward. What amount of prism is induced and what is the base orientation?	
c.	What is strums' of conoid?	
d.	Define back vertex power and write down its formula.	
e.	Define approximate power and equivalent power.	
f.	Write down range of visible spectrum and UV light.	
g.	What is the meaning of refraction and total internal reflection?	
h.	Explain resultant vertical prismatic effects.	
i.	Describe spherical lens form.	
j.	Do the Simple transposition 1.+2.25/-3.50×95, 2.-7.75/-8.00×87, 3.+4.25/-5.75×75, 4.-13.50/+2.50×11	
k.	What is rotary prism?	

SECTION II

Q.1.	Short Notes (4 out of 5)	(4 × 5 = 20)
a.	+5.00/+1.00×95, Base curve=+6.00. Find out the fraction value using toric transposition steps.	
b.	Write a note on spherical equivalent with example.	
c.	Explain in detail about spherical aberration.	
d.	A lens has the following parameters. F1=+11.00D, F2= -6.00D, n=1.532, tp=1.0mm, lens diameter=50mm. find out tc using exact sagitta formula.	
e.	Write in detail about Fresnel Press-on prism.	

Q.2.	Essay (1 out of 2)	(1 × 10 = 10)
a.	1. Derive sag formula with proper diagram. 2. A certain lens of index 1.586 has a convex spherical front surface. The sag of the front surface is 1mm for a chord of whose length is 22mm. What is the power of the front surface?	
b.	Explain in detail about Prentice's rule with example.	